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Quantitative text analysis – Essex Summer School

LLMs as data annotators

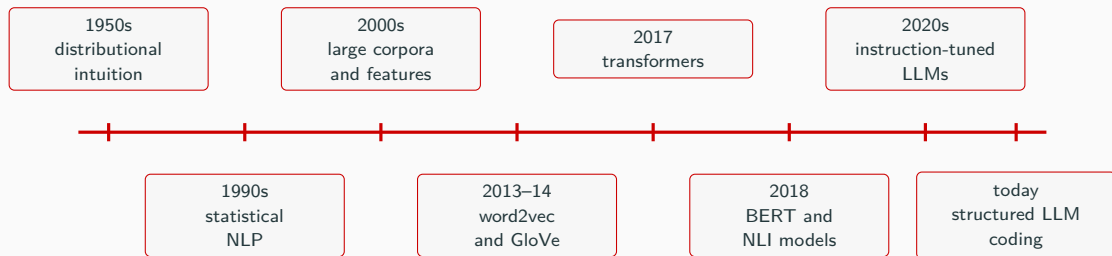
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Today's class

- How can we use LLMs in social science research?
- Three themes:
 - **coding**: annotation, classification, extraction
 - **validation**: checking whether labels measure the intended concept
 - **reporting**: preserving the full audit trail
- Lab session: LLM-assisted coding

A short NLP timeline



NLP moved from hand-built features and task-specific classifiers toward reusable language models that can represent and generate language.

Encoder and decoder models

Encoder-only BERT

An encoder model (like BERT) is like a **careful reader**.

It takes a text, builds a rich understanding of it, and then passes that understanding to a task-specific head.

Typical question:

“Given this text, which label fits best?”

Examples:

- Is this paragraph about the economy?
- Is this sentence positive or negative?

Prompt-based generative models

A generative model is like a **reader-writer**.

It reads the prompt and then produces the next words that best continue the interaction.

Typical question:

“Given these instructions and this text, what should the answer be?”

Examples:

- Summarise this argument.
- Code this speech using the codebook.

What changes for applied text analysis?

	BERT-style workflow	Prompt-based generative workflow
Input	text	text plus instructions
Model output	representation for a classifier	answer in a requested format
Best fit	stable labels and many similar cases	interpretive tasks with codebook rules
Main risk	task-specific training still needed	prompt choices become measurement choices

Main shift: the task definition moves closer to the model input. That makes the method flexible, but also makes prompt design part of the method.

Why LLMs are tempting for QTA

LLMs are attractive when the coding task is hard to reduce to a keyword, a dictionary, or a single labelled training set.

- They can apply natural-language codebooks and return structured labels with short evidence statements.

But: an LLM-coded dataset is still a measurement instrument. It needs a codebook, validation, and reporting.

Bridging qualitative and quantitative text analysis

Traditional QTA often asks researchers to translate rich concepts into simple, scalable features.

Qualitative coding	Quantitative text analysis
close reading	large corpora
context-sensitive labels	comparable variables
nuanced codebooks	reproducible workflows
careful interpretation	statistical summaries

LLMs can help bridge this divide by applying **qualitative categories at scale**.

The challenge is to keep the qualitative discipline while scaling up.

Balluff et al. (2026): trade-off mindset

- Balluff and colleagues warn against unreflective LLM adoption in research: newer and larger does not automatically mean better for a given research task.

Trade-off mindset

Ask what is gained, what is lost, and whether a simpler method would be more transparent for the same research purpose.

- They argue to use LLMs when they are justified by the task, not because they are available.

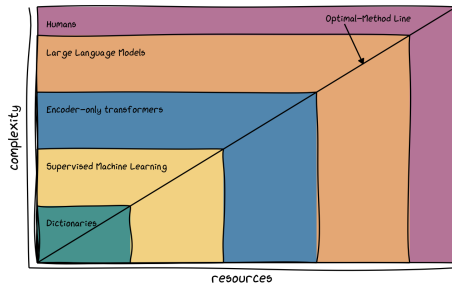


Figure 1. Mental model sketch to visualize the trade-off function to consider when choosing tools from the text-as-data toolbox. The plot shows each method's capability range: the vertical span indicates the task complexity each method can handle, while the horizontal position shows the resources required to implement that method for tasks within its capability. The areas overlap because more sophisticated methods can also solve simpler tasks: LLMs can do everything dictionaries can do, just at higher cost. The "optimal method line" represents the most efficient choice: the least resource-intensive method capable of handling a given task complexity.

Source: Balluff et al. (2026).

McClaren and colleagues ask: if we use an LLM as a coder, how much do the results depend on the choices we make around it?

Choice	What they find
Which model?	larger models are not always better
How many examples in the prompt?	examples sometimes help, sometimes do not
How the prompt is written?	personas and reasoning prompts can help, hurt, or make little difference
Which task?	the best setup varies across coding tasks

Takeaway: LLM coding is not one method but a pipeline. **Model, prompt, examples, and task definition all need to be justified and tested.**

The prompt as a measurement instrument

Main idea

For LLM coding, the prompt is part of the measurement instrument.

- It defines what the model is asked to see in the text, how edge cases should be handled, and what evidence should be returned.
- Prompt choices can therefore be thought of as **researcher degrees of freedom**.

Lab 10 prompt structure

Role

Careful qualitative coder of UK parliamentary speeches.

Task

Code each speech for **credit claiming** and **blame attribution**.

Decision rules

Use only the speech text. Do not infer from party identity or outside knowledge. Code **mixed** when both moves are present.

Output

Return fixed fields: credit, blame, target actor, policy area, confidence, and evidence.

Uncertainty

Use lower confidence when the evidence is ambiguous; choose **unclear_or_none** when no actor is clearly credited or blamed.

Validation-first workflow (McClaren et al. 2026)

Step 1	Step 2	Step 3
define constraints	compare models on your validation sample	tune prompts only if needed
hardware, budget, privacy, time	use simple zero-shot prompts first	examples, personas, reasoning prompts

Intuition

Do not spend time polishing a weak model before you know whether it is the right model for the task.

Why quallmer?

- If LLM coding is a measurement pipeline, we need software that stores more than a prompt and a spreadsheet of labels
- Lab 10 uses quallmer for this workflow, which helps organise:
 - codebooks
 - structured coding runs
 - comparisons
 - validation
 - provenance trails



Codebook-centred LLM coding

- `quallmer` defines the research workflow
- `ellmer` provides the model interface and structured-output tools
- Providers can include OpenAI, Anthropic, Google, or local Ollama models
- Live coding requires a provider setup and often an API key



Running example: leader speeches

Lab 10 uses the House of Commons leader corpus. Which Labour and Conservative leaders are more likely to claim credit, and which are more likely to attribute blame?

Credit claiming

A speaker claims that they, their party, their government, or another actor delivered a positive outcome, improvement, achievement, or successful policy.

Blame attribution

A speaker attributes a negative outcome, failure, harm, or public problem to an actor.

- The task is interpretive, but the resulting labels can be compared across leaders, parties, and periods.

Prompt component: task role

- Use the role to clarify the kind of task, not to prime a preferred interpretation.

A safer role names the coding practice and the domain: for example, “careful qualitative coder of UK parliamentary speeches.”

A riskier role smuggles in a substantive expectation: for example, “expert detecting government failure” or “partisan analyst of political blame.”

McClaren et al. (2026): prompt style choices can matter unpredictably, so persona-style roles should be treated as **sensitivity checks rather than neutral defaults**.

Prompt component: category definitions

Category	Definition
credit claiming	an actor is linked to a positive outcome, achievement, delivery, or improvement
blame attribution	an actor is linked to a negative outcome, failure, harm, or unresolved problem
mixed	both credit claiming and blame attribution are clearly present
neither	no clear credit or blame move is made

Definitions should be close to the concept, not just lists of keywords. A speech can contain “jobs” or “failure” without making a credit or blame claim.

Prompt component: decision rules

Rule	Why it matters
use only the speech text	avoids importing outside knowledge about parties or leaders
identify the actor being credited or blamed	turns a vague label into an interpretable claim
keep a short evidence quote	makes decisions auditable

- Decision rules tell the model what to do when a speech is messy, indirect, or mixed.

Prompt component: output schema

A structured output schema tells the model to return the same fields for every speech. In practice, those fields become columns in a data frame: **one row per speech, one column per code.**

Column	Type
credit_claim	TRUE/FALSE
blame_attribution	TRUE/FALSE
main_claim_type	fixed category
target_actor	fixed category
policy_area	fixed category
confidence	number
evidence	text

Key point: the schema turns model output into a rectangular dataset that can be analysed and validated.

Prompt component: confidence

Prompt component: uncertainty

LLMs often produce definite labels even when the text provides only weak or ambiguous evidence.

Ask for an ordinal confidence judgement

After assigning a label, require the model to report confidence as **low**, **medium**, or **high**, based only on the evidence in the text.

Example prompt instruction

Assign one label and one confidence level: `low`, `medium`, or `high`. Base confidence on the clarity and strength of the textual evidence, not on your general knowledge. If no label is sufficiently supported, select `unclear`.

Validation evidence



Anchor

The human-coded validation sample is the reference point for assessing whether the model is measuring the intended concept.

Validation question 1: content validity

Does the instrument cover the concept?

For credit/blame coding, the prompt should cover positive outcomes, failures, mixed claims, actor targets, and evidence.

This is a codebook question first.

Validation question 2: criterion validity

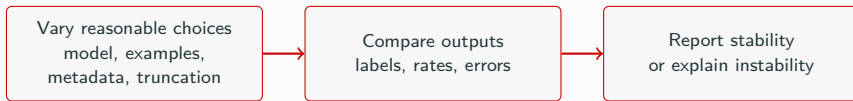
Human judgement as anchor

Criterion validation asks whether model labels agree with defensible human labels on a validation sample.

That sample should include all categories, borderline cases, different speakers and periods, and examples where the model is likely to fail.

Rule: do not validate only on easy cases.

Validation question 3: stability



Core question

Would the substantive conclusion survive reasonable changes to the pipeline?

Error analysis

Accuracy is a starting point for validation.

Model label	Human label	
	claim	no claim
claim	agreement	false positive
no claim	false negative	agreement

Then inspect whether errors cluster by party, leader, period, agenda, or unsupported evidence.

What to report

Instrument

concept, codebook, fields

Model pipeline

provider, version, settings

Validation

human sample, errors

Interpretation

uncertainty, limits

Concluding remarks

- LLMs can help scale qualitative judgement, but they do not replace measurement design.
- The strongest projects combine clear concepts, transparent codebooks, validation against human judgement, and honest reporting of uncertainty.